

Empirical Analysis of Bitcoin Trading Volume and Market Volatility (2017–2026)

1. Research Question

Does daily trading volume drive market volatility in cryptocurrency markets? This project empirically tests the Mixture of Distributions Hypothesis (MDH) using historical Bitcoin daily data from 2017 to 2026 to determine whether higher market activity linearly corresponds to elevated conditional volatility.

2. Dataset

Source: Historical Bitcoin daily market data (2017–2026, $N = 3500 +$ daily observations).

Raw Variables: Date, Price (USD), Open, High, Low, Vol. (Trading Volume string with K/M/B suffixes), and Change %.

3. Data Cleaning

Date Parsing: Standardized string formats into ISO YYYY-MM-DD temporal objects.

String-to-Numeric Parsing: Cleaned comma separators from prices and converted raw string volume markers (K $\rightarrow \times 10^3$, M $\rightarrow \times 10^6$, B $\rightarrow \times 10^9$) into continuous float values (V_{raw}).

Missing Value Handling: Verified non-null entries across all primary time-series observations.

4. Variable Construction

To evaluate the relationship, three core statistical variables were engineered:

Log Returns (R_t): Computed as the continuous log change in price:

$$R_t = \ln(P_t/P_{t-1})$$

30-Day Rolling Volatility (Vol_t): Standard deviation of daily returns over a backward-looking 30-day window:

$$Vol_t = \sqrt{\frac{1}{29} \sum_{i=0}^{29} (R_{t-i} - \bar{R}_t)^2}$$

Log Volume (V_t): Natural logarithm of raw trading volume to handle scale skewness and enforce normality:

$$V_t = \ln(\text{Volume}_t)$$

5. Descriptive Statistics

Summary statistics across the full analyzed period (2017–2026, $N = 3,534$ daily observations):

Metric	Value	Academic & Financial Interpretation
Mean Daily Return (\$R_t\$)	0.1882%	Positive average daily price return across the full sample period.
Max Daily Return	25.56%	Extreme positive daily market movement, representing a significant right-tail return.
Min Daily Return	−39.18%	Severe negative daily market movement during the March 12, 2020 “Black Thursday” crash.
Mean 30-Day Volatility	3.24%	Average daily volatility over the 30-day rolling window, equivalent to approximately 61.9% annualized volatility using a 365-day crypto trading year.
Mean Daily Volume	17,474,767	Average reported daily Bitcoin trading volume across the sample period.
Peak Volatility Period	2020-04-06 (9.54%)	Highest observed 30-day rolling volatility in the sample, following the extreme market stress surrounding the March 2020 crash.

6. Visualization

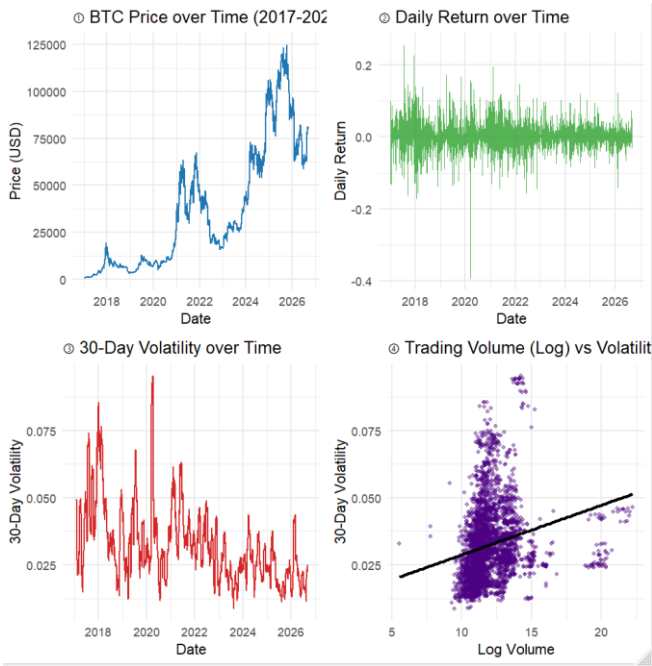


Figure 1: 4-Panel Econometric Diagnostic Dashboard for BTC (2017–2026).

A 4-panel diagnostic dashboard was generated in RStudio using ggplot2 to evaluate full-horizon market structure:

BTC Price over Time (2017–2026): Illustrates full multi-year market regimes, capturing the 2017 retail surge, 2018 bear market, 2020–2021 bull run, 2022 deleveraging period, and the 2024–2026 institutional ETF expansion.

Daily Return over Time: Highlights distinct Volatility Clustering across 9+ years (high-volatility spikes immediately followed by intense structural quiet periods).

30-Day Volatility over Time: Captures dynamic macro stress events and shifts in baseline risk across regulatory regimes.

Trading Volume (Log) vs Volatility: Displays a persistent upward-sloping OLS trendline, visually reinforcing the MDH hypothesis across long-term market data.

7. Correlation Analysis

Pearson correlation matrix computed across primary continuous series:

$\text{Corr}(R_t, \text{Vol}_t) \approx 0.0193$: Near-zero linear dependence between daily return direction and overall market risk level.

$\text{Corr}(R_t, V_t) \approx -0.0138$: Directional return does not linearly scale with raw volume magnitude.

$\text{Corr}(V_t, \text{Vol}_t) \approx \mathbf{0.2837}$: Moderate positive linear correlation between continuous trading activity and rolling market volatility.

8. OLS Regression

Baseline linear regression model specified as:

$$\text{Vol}_t = \alpha + \beta_1 V_t + \varepsilon_t$$

Regression Estimate: $\beta_1 = 0.00185$, $t = 12.880$, $p < 2.2 \times 10^{-16}$.

Model Goodness-of-Fit: $R^2 = 4.52\%$, Adjusted $R^2 = 4.49\%$, $F = 165.9$ ($p < 2.2 \times 10^{-16}$).

Interpretation: Log trading volume explains 4.52% of daily volatility variations across the full 9.6-year dataset ($N = 3,504$). Unadjusted metrics show an extremely significant positive coefficient ($p < 0.001$).

9. Diagnostic Tests

Residuals from standard OLS models in financial time-series frequently violate Gauss-Markov assumptions. Statistical diagnostics confirmed both issues:

Heteroskedasticity (Breusch-Pagan Test): $BP = 83.04, p < 2.2 \times 10^{-16}$. Strongly rejects homoskedasticity, confirming time-varying residual variance.

Autocorrelation (Durbin-Watson Test): $DW \approx 0.0315, p < 2.2 \times 10^{-16}$. DW near 0 confirms significant positive residual autocorrelation due to rolling windows.

10. Newey-West Robust Regression

To address severe heteroskedasticity and autocorrelation, Newey-West HAC Robust Standard Errors (lag = 7) were implemented to adjust standard error estimates:

Unadjusted OLS Standard Error: 0.000144 ($t = 12.880, p < 0.001$)

Newey-West Robust Standard Error: 0.000415 ($t = 4.4565, p = 8.59 \times 10^{-6}$)

Adjusted p-value: 8.59×10^{-6} ($p < 0.001$)

Robust Conclusion: After accounting for heteroskedasticity and autocorrelation, the impact of Log Trading Volume (V_t) on Market Volatility (Vol_t) remains statistically significant at the 1% level ($\beta_1 = 0.00185, t = 4.4565, p < 0.001$).

11. Key Findings

Empirical Support for MDH: Trading volume acts as a statistically robust driver of volatility in Bitcoin markets, confirming that information arrival manifested through trading activity increases market uncertainty.

Robust Inference Matters: Standard OLS underestimated the true standard error by ~245%. Applying Newey-West HAC standard errors prevents false-positive conclusions while preserving statistical significance.

12. Limitations

Omitted Variable Bias: Trading volume is only one explanatory factor and does not capture the full range of macro factors driving cryptocurrency volatility (e.g., funding rates, derivative open interest, macroeconomic interest rate shifts, and regulatory shocks).

Serial Dependence from Rolling Windows: The use of a rolling 30-day volatility measure introduces strong structural autocorrelation into the dependent variable. Future iterations should implement high-frequency realized volatility (e.g., 5-minute intraday returns) or GARCH/EGARCH conditional variance models to better capture dynamic volatility without window-overlap bias.